

## Scalability & Future Plan

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|---------------|--------------|
| Date          | 1 July 2026  |
| Team ID       | XXX          |
| Project Name  | Smart Lender |
| Maximum Marks | 1 Mark       |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                    | Impact                                         | Priority to Address<br>(High / Medium / Low) |
|------|-------------------------------|------------------------------------------------|----------------------------------------------|
| 1    | Single-node Flask deployment  | Cannot handle high concurrent user traffic     | High                                         |
| 2    | Manual model retraining       | Delays incorporating new applicant data trends | Medium                                       |
| 3    | SQLite database for user data | Not suited for large-scale concurrent writes   | Medium                                       |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                        | Proposed Upgrade / Solution                                                               |
|------|--------------------|--------------------------------------|-------------------------------------------------------------------------------------------|
| 1    | User Load          | Single Flask server, low concurrency | Deploy on cloud (IBM Cloud/AWS) with load balancer and multiple Gunicorn worker instances |
| 2    | Data Storage       | SQLite local database                | Migrate to a managed PostgreSQL/MySQL cloud database                                      |
| 3    | Performance        | Synchronous single-model inference   | Add model caching, batch inference and async request handling                             |
| 4    | Security           | Basic password hashing               | Add JWT-based auth, HTTPS, stricter input validation and rate limiting                    |

### Future Roadmap:

| Phase   | Planned Feature / Enhancement       | Target Timeline | Expected Impact                   |
|---------|-------------------------------------|-----------------|-----------------------------------|
| Phase 1 | Cloud deployment & load balancing   | Q3 2026         | Support 1000+ concurrent users    |
| Phase 2 | Migrate to PostgreSQL & add caching | Q4 2026         | Improved data reliability & speed |
| Phase 3 | Mobile app & public REST API        | Q1 2027         | Wider accessibility for users     |